

23AID303 - Quantum Computing and Algorithms

Lab 2 Exercises

1. Apply a Hadamard gate to a qubit initially in state $|0\rangle$. What is the probability of measuring $|1\rangle$? Run the simulation in Qiskit and verify.
2. Create a quantum circuit where a qubit is flipped only when the control qubit is in state $|1\rangle$. Which gate do you use? Simulate.
3. Apply X then H to $|0\rangle$. What is the outcome you observe?
4. Apply $H \rightarrow Z \rightarrow H$ to $|0\rangle$. What is the final state? Verify with statevector.

Note: from qiskit.quantum_info import Statevector

```
state = Statevector.from_instruction(qc)
```

will collect the state vector, which can be printed.

5. Apply 4 T gates in sequence to $|1\rangle$. What gate does that effectively simulate? (Hint: Z)
6. Apply a combination: $H \rightarrow T \rightarrow H$ to $|0\rangle$. What gate does this simulate? Can you express it in terms of X/Y/Z?
7. Try all single-qubit gates (I, X, Y, Z, H, S, T) on $|0\rangle$ and $|1\rangle$. Verify with Statevector.
8. Use Qiskit to apply an X gate followed by a Z gate to $|0\rangle$. What is the resulting state vector?
9. Apply two qubit gates on $|0\rangle$ and $|1\rangle$. Check the results.
10. Experiment three qubit gates on $|0\rangle$ and $|1\rangle$. Validate the results.